



Lecture 03 September 2026

Philosophical fundamentals of AI

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Part 1

Practitioner foundations

Learning and reasoning

Trustworthy AI

Logic interlude

Part 2

Philosophical foundations for
intelligent machines

Conditions on intelligence

Conditions on personhood

Part I: Practitioner foundations (Chapter 1 of CHLEP)

AI = (designed) learning + reasoning **machines**

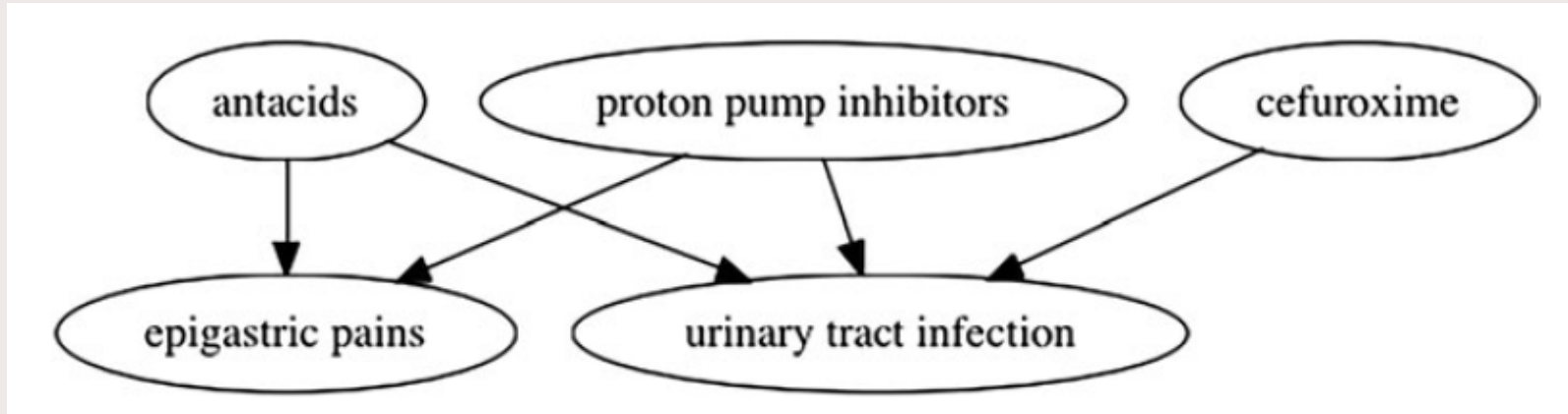
Learning = getting better at a *task* through *experience*

- Requires a *metric* that measures improvement (better/ worse)
- Is always relative to a specific task

Reasoning = logical deduction, probabilistic inference, inference to the best explanation

(“manipulation of previously acquired knowledge to answer a new question”)

Probabilistic inference



Where does knowledge come from?

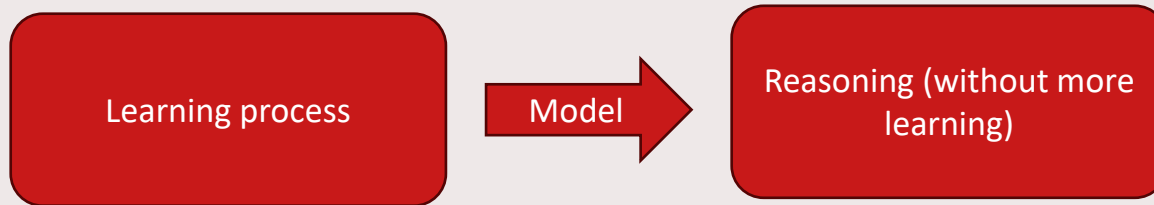
Philosophically, a system's domain knowledge can be:

- (a) a priori --- derived independently of experience (e.g., mathematical and logical truths; “constructed knowledge” of ethical guidelines)
- (b) testimonial --- derived from (statements of) domain expert knowledge
- (c) empirical --- derived from experience (e.g., learning from data)

Philosophically, **knowledge that P** consists of a *warranted true belief* in P, where belief can be understood as a representation that P relied upon by the system, and warrant consists of P's being acquired through a non-accidentally reliable process in normal conditions.

Learning from experience

A truly intelligent being is able to learn *from any experience*, not merely apply a process that *resulted* from learning in the past. “One of the grand challenges of AI is achieving a truly integrated learning and reasoning mechanism” (CHLEP p. 17).



Explainability and Trustworthiness

The TAILOR Handbook of Trustworthy AI

An encyclopedia of the major scientific and technical terms related to Trustworthy Artificial Intelligence

According the [Ethics Guidelines for Trustworthy AI](#) by High-Level Expert Group on Artificial Intelligence, **Trustworthy Artificial Intelligence (Trustworthy AI)** has three components, which should be met throughout the system's entire life cycle. Indeed, it should be:

1. *lawful*, complying with all applicable laws and regulations;
2. *ethical*, ensuring adherence to ethical principles and values;
3. *robust*, both from a technical and social perspective since, even with good intentions, AI systems can cause unintentional harm.

Framework for Trustworthy AI

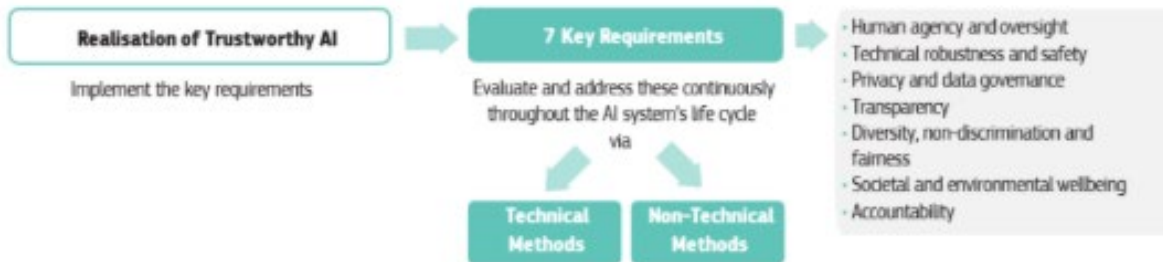
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High-Level Expert Group
on Artificial Intelligence. Ethics Guidelines for
Trustworthy AI. 2019. URL: [https://digital-
strategy.ec.europa.eu/en/library/ethics-
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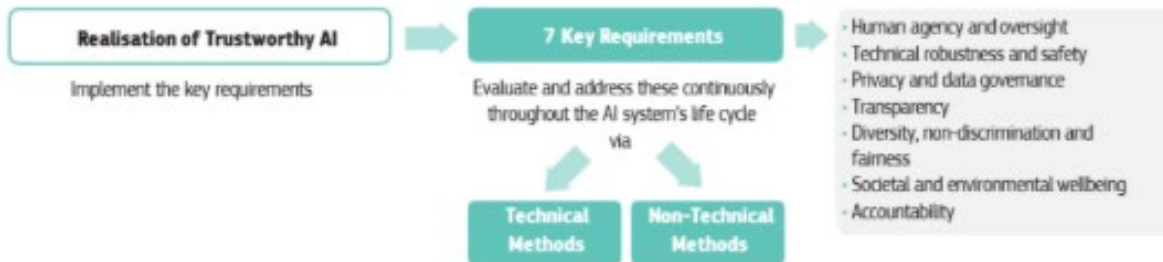
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Explicability: Interpretable or Explainable AI

“On top of the mere output of a machine learning or reasoning algorithm, we ... need techniques to explain these outputs.”

Interpretable AI: reasoning is evident from the model (e.g., a decision tree)

Explainable AI: “mechanisms to inspect black box models (e.g., neural networks)”

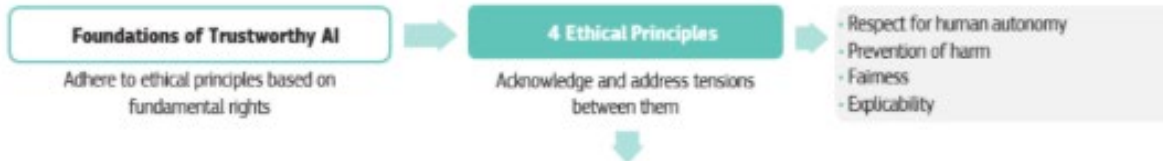
- Model-specific vs. model-agnostic
- Examples are **feature importance** and **counterfactual explanations**

Framework for Trustworthy AI

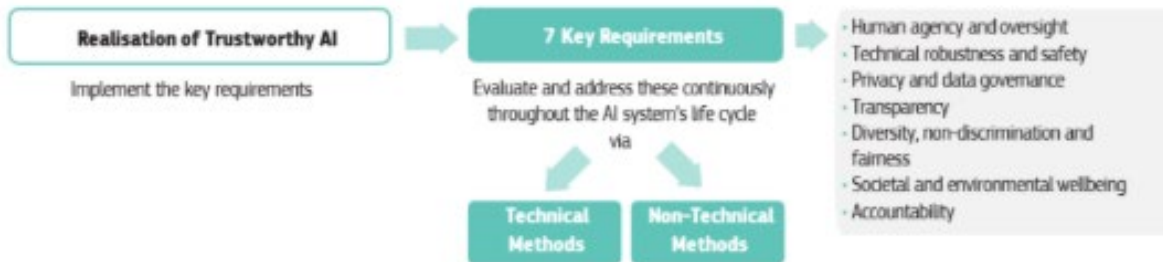
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Robust AI

Robustness is the property of an AI model, such that its reasoning will continue to obtain for new data.

For example, suppose a model is trained to identify a given kind of cancer from an image (e.g., of adult scans).

- Will it show the same accuracy for new images of the same kind, from the clinical setting?
- Will it show the same (or sufficient) accuracy for images that are known to be different (e.g, scans from children)?
- Is it difficult to create adversarial images or queries?

A robust model leads to affirmative answers to these questions.

Interlude: Informal logic and ethical reasoning

- An **argument** is a set of propositions containing at least one premise and one conclusion. Premises are intended to support the conclusion (i.e., to show that the conclusion is true).
- 1) Either I am talking too fast, or it's likely I won't get through all my slides during the lecture.
 - 2) I am not talking too fast.
 - 3) Therefore it's likely I won't get through all my slides during the lecture.

Deductive Arguments and Validity

- **Validity** is a desired feature of a deductive argument. Validity is that property of arguments such that **if the premises are true, then the conclusion is guaranteed to be true.**
 - 1) All toads are mammals.
 - 2) Socrates was a toad.
 - 3) Therefore, Socrates was a mammal.

The idea is that such an argument form is “truth-preserving” or “knowledge-preserving”.

Deductive Arguments and Soundness

- A second essential feature of an argument is that its **premises are true**.
 - When an argument is valid and has all true premises, then we call it **sound**.
- 1) All humans are mammals.
 - 2) Socrates was a human.
 - 3) Therefore, Socrates was a mammal.

Is-Ought Gaps

“No ought-judgment may be correctly inferred from a set of premises expressed only in terms of ‘is’.”

(“Hume’s Moral Philosophy,” *Stanford Encyclopedia of Philosophy*, Accessed 4 September 2016)

To illustrate:

- 1) Robot type A is square.**
- 2) Therefore, we ought to build a robot of type A.**

Simple normative arguments

- Stepping on a dog's tail for fun causes harm.
- It is wrong to cause harm, other things equal. (Non-maleficence)
- Therefore, it is wrong to step on a dog's tail for fun.

Inductive Arguments

- “Valid arguments are risk-free. Inductive logic studies risky arguments” (Hacking 2001, p. 11).
- 1) These four oranges taken at random from the box are good.
 - 2) Therefore, the oranges in this box are good.

Arguments using probabilistic reasoning (e.g., Bayesian reasoning) and then stating a non-probabilistic conclusion are risky in this way.

Normative argumentation involving rights

- EU's AI Act fails to set gold standard for human rights - European Digital Rights (EDRI)

Example: “The AI Act falls short of civil society’s demand to ensure that EU-based AI providers whose systems impact people outside of the EU are subject to the same requirements as those inside the EU. The Act does not stop EU-based companies from exporting AI systems which are banned in the EU, therefore creating a huge risk of violating rights of people in non-EU countries by EU-made technologies that are essentially incompatible with human rights. Additionally, the Act does not require exported high-risk systems to follow the technical, transparency or other safeguards otherwise required when AI systems are intended for use within the EU, again risking the violation of rights of people outside of the EU by EU-made technologies.”

Extraction of argumentation from text

- Policy P in jurisdiction 1 leaves in place a significant risk that human rights will be violated in jurisdiction 2, compared to alternative policy P*
- (Other things equal, policies should reduce the risk of human rights violations in extraterritorial jurisdictions, relative to alternatives.)
- Therefore, policy P is inadequate in jurisdiction 1.

Part II: Philosophical foundations for intelligent machines

Philosophical fundamentals: terminology and agenda-setting

Necessary and sufficient conditions

Conditions on intelligence

Conditions on personhood

The Dartmouth workshop (1956)

Dartmouth workshop.



First golden age of AI:
reasoning-based, expert
systems



AI winter(s)

Current golden age of AI:
machine learning, LLMs

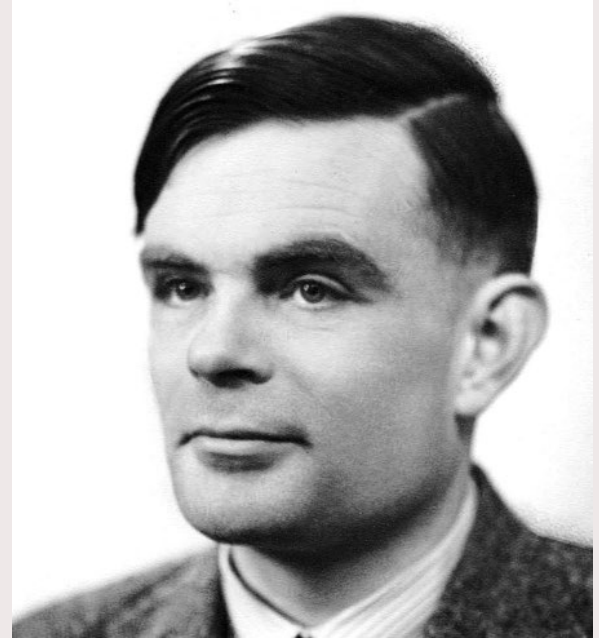
Some foundational aims

Classical AI aims to build artificial machines that have intelligence. It holds that any feature of (human) intelligence can be built into a machine.

In addition, it holds that a machine so built would help us understand (human) intelligence.

Technical AI is a set of computer-science methods for machine learning and reasoning.

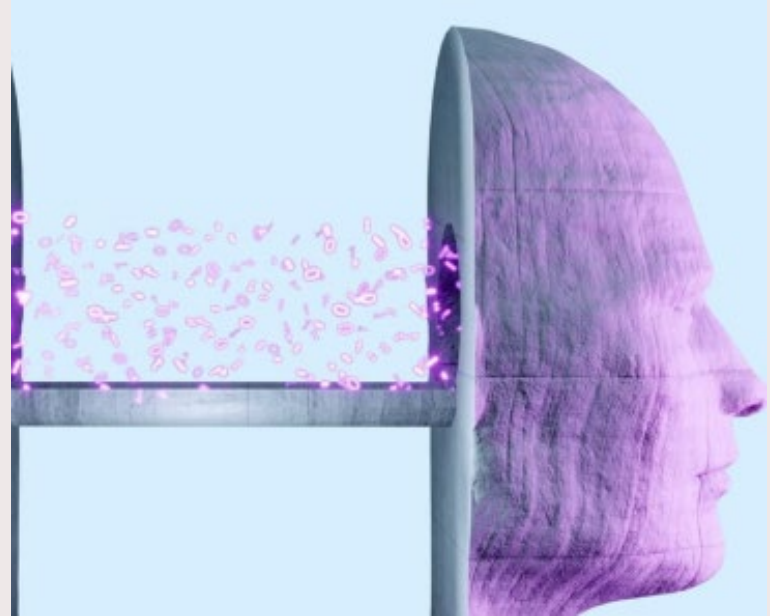
Strong AI actually has a mind and cognition.



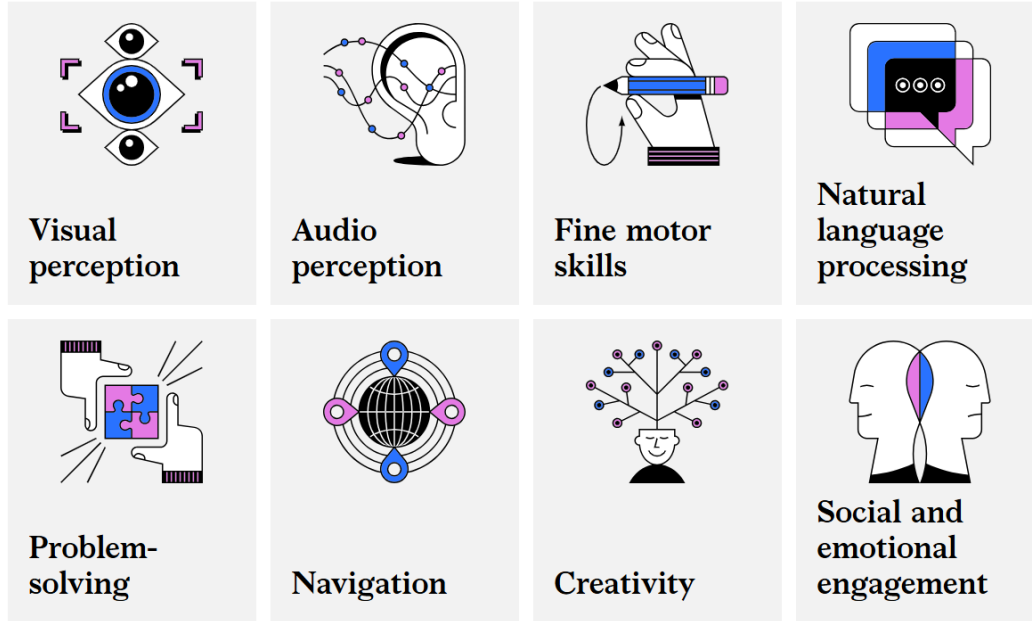
Artificial General Intelligence

We can have the ambition to recreate human or superhuman intelligence (Classical AI), while maintaining the current methodology of Technical AI (with which you are becoming familiar in your program).

Image: [What is Artificial General Intelligence \(AGI\)? | McKinsey](#)



McKinsey claims these skills are needed for AGI



Functional definition of intelligence

The Turing Test (1950): If a human cannot tell a human from a machine in a blinded test involving a conversation with an unknown entity (an “imitation game”), then the machine is intelligent, i.e., it thinks.

But: the test cannot be a good definition of intelligence, because there could be intelligent beings that cannot pass the test (of imitating human written communication).

Goal-oriented (instrumental) definition of intelligence

Intelligence is a capacity to achieve goals flexibly and in a wide range of conditions.

To some extent, it is system dependent or a systemic capacity.

By itself, this definition does not say what goals are “intelligent” to pursue, e.g., self-preservation.



Philosophy of AI

What does it take for computation to be intelligent (“to think”)? AI has prompted philosophers, cognitive scientists, and computer scientists to consider carefully what we mean by such terms. Some computation does not involve acting on the world, it is “passive” except insofar as it generates an output and communicates it via an interface. Can that be intelligence?

Arguments that intelligence is computation-plus

- The 4E argument: only embodied, embedded, enactive, and extended computation can be intelligent
- The “Chinese room” argument: only computation with intentionality can be intelligent

Chinese room thought experiment visualized



[The Chinese Room Experiment](#) | [The Hunt for AI](#) | [BBC Studios](#) -

[Y29Tube](#) [Title]

Structure of argumentation

Necessary conditions/ modus tollens

1. E is strongly intelligent only if E has some capacity C (e.g., of being able to intervene in the world). (Necessary condition for intelligence)
2. AI does not have C.
3. Therefore, AI is not strongly intelligent.

Structure of argumentation

Sufficient conditions/ modus ponens

1. If E has C, E is intelligent (Sufficient condition for personhood).
2. AI has such-and-such a capacity.
3. Therefore, AI is intelligent.

Candidate necessary conditions of intelligence

Language use

Flexible goal-
achievement

Rational choice

Enactive
engagement



If any one of these is a necessary condition of intelligence, but impossible for an intelligent, silicon-based machine to possess, then that machine cannot be intelligent.

Candidate necessary conditions of personhood

Consciousness

Selfhood

Responsibility

Free will



If any one of these is a necessary condition of personhood, but impossible for an intelligent, silicon-based machine to possess, then that machine cannot be a person. Usually, being a person is considered a basic kind of moral status (high degree of moral value requiring respect/ possessing rights).

Some questions to keep in mind for this course:

Can every human activity be broken down into tasks? What are ethics tasks?

Are there any metrics we can use to measure success at ethics tasks?

Is being good at ethics a component part of high intelligence?

Is deductive reasoning a necessary part of high intelligence?

That's it! Now on to some practical things...

Client consult meetings

First round of coaching meetings: format

Group and topic sign-up

Client consult meetings

A short introduction from the client

30-40 minutes for any student questions about the topic

Possible questions you have might concern:

- Target context (literature or website suggestions)
- Data sharing
- Prompting and technical suggestions
- Ethical issues

First Round of Coaching Meetings: Format

Timeframe and participants: 30 minutes total, two groups, 8 students + instructor

Prepare to answer questions about the concrete results of literature search, initial exploration of technology options, initial exploration of target context and projected user. Each individual student should be prepared to say something about what they have been doing to get ready for the submission of Phase 1.

Give initial answers to the following questions:

- What is the problem or need to which the AI tool is meant to respond?
- What frameworks or concepts can be used to understand the problem and structure a response to it?
- What technical options appear available to respond to the problem or need?

Self-assessment for coaching meetings

Basic	Intermediate	Advanced
Problem or need comes from imagination or brainstorm	Problem or need comes from literature or real life	Problem or need has been refined on the basis of literature or real-life sources
Framework/ concept is not present or merely intuitive	Framework/ concept comes from literature	Framework/ concept has been refined on the basis of literature
Technical options are explored at a layperson level	Technical options are explored at a scientifically informed level	Scientifically-informed options are explored in a systematic way

Sign-up for groups and topics